IncidencePrevalence

Estimate Incidence Rates and Prevalence in OMOP CDM

2026-06-25

Oveview

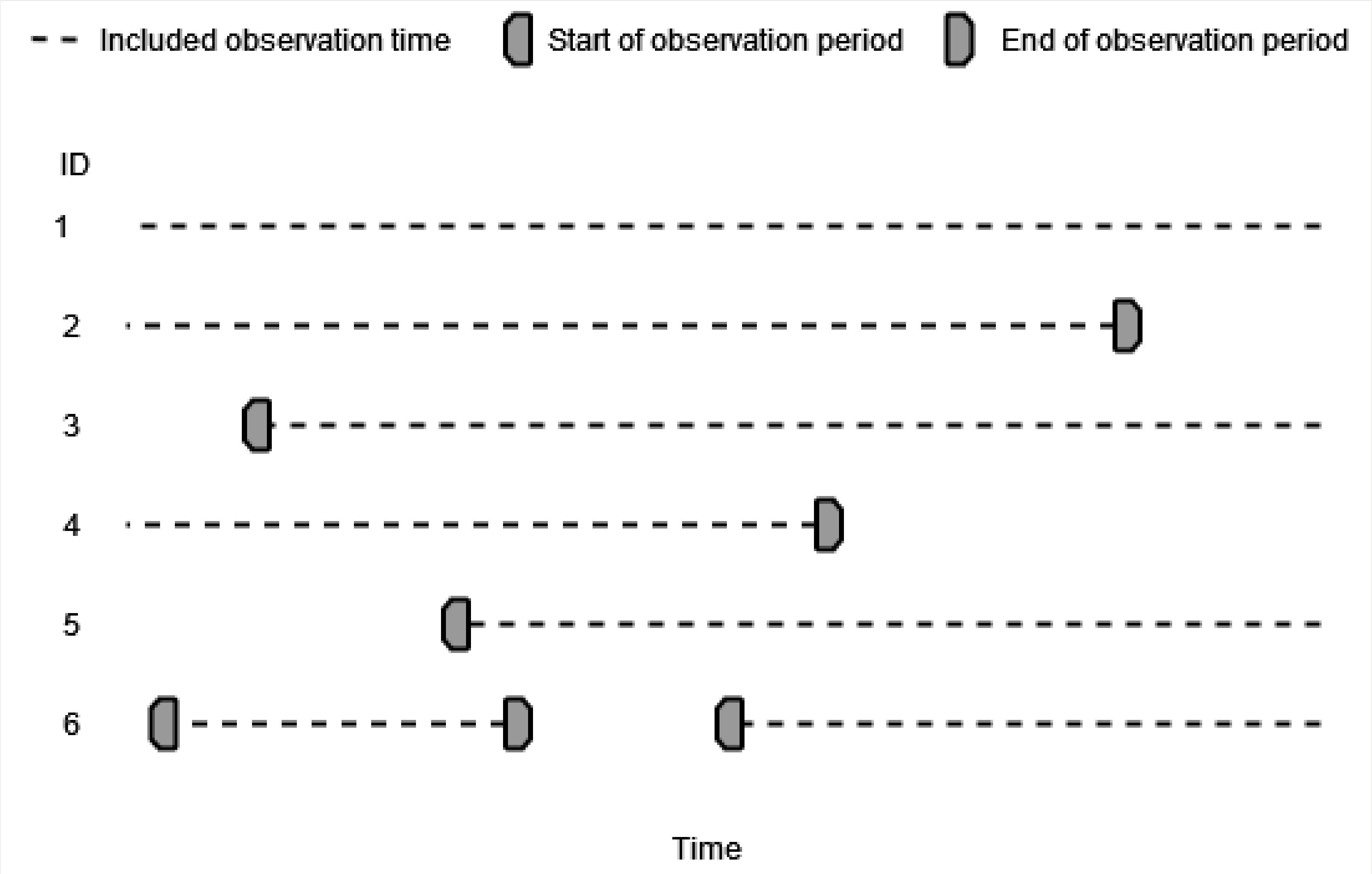
- **Concepts**
- Interface
- More information

Concepts

1. Denominator population
2. Incidence rates
3. Prevalence
    - Point prevalence
    - Period prevalence

Denominator population

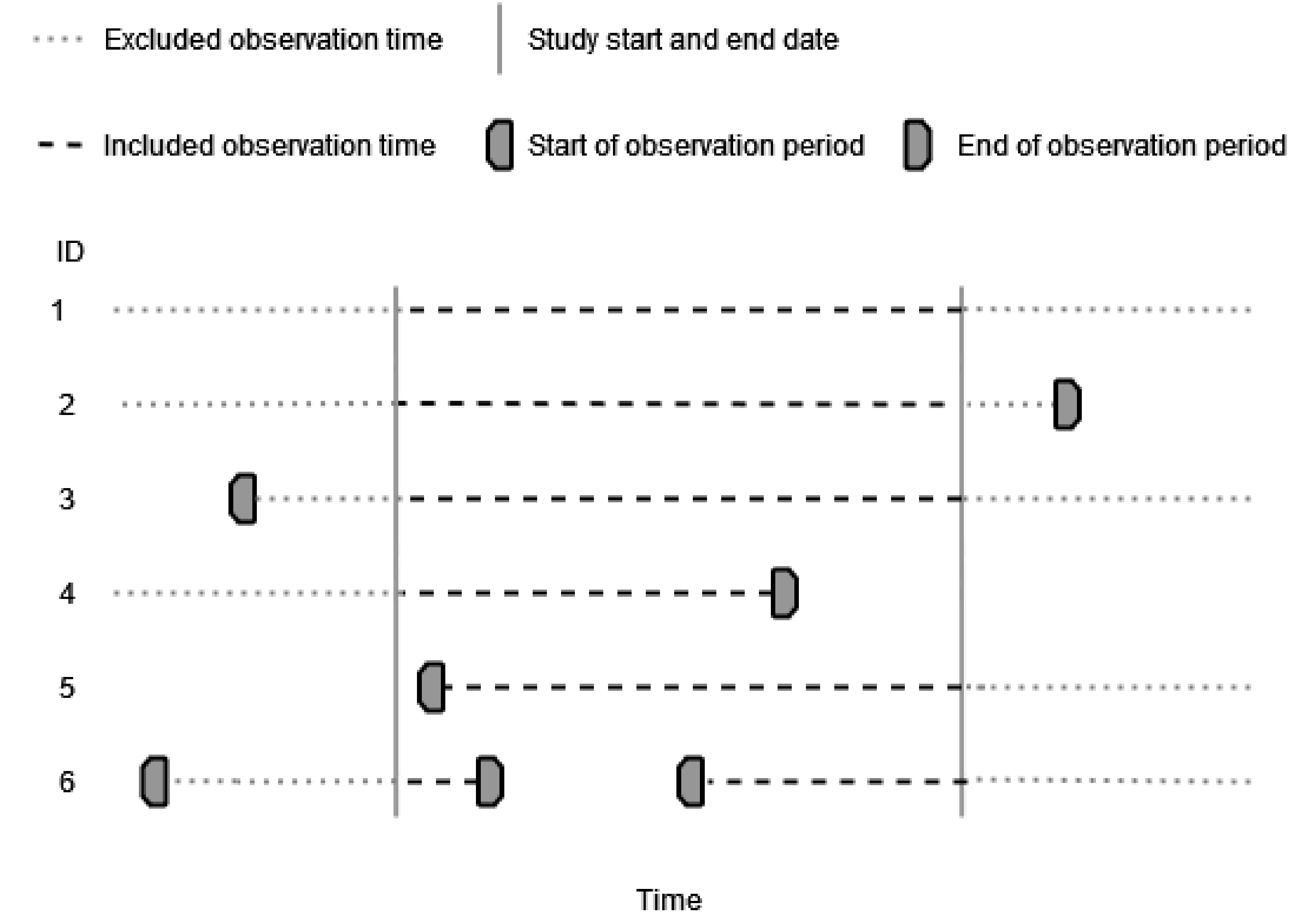
Observation periods



⬅ Back to Presentations

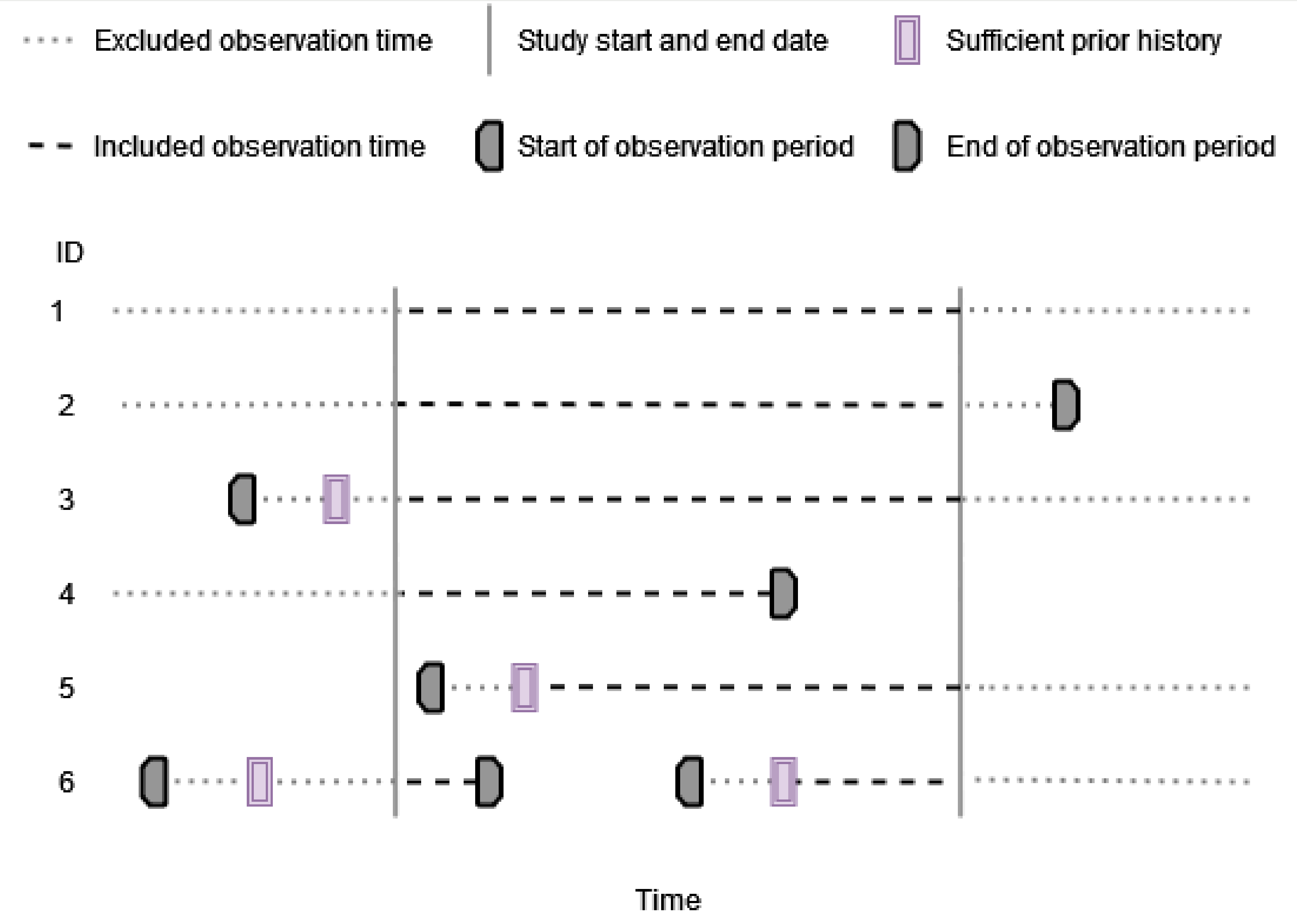
Denominator population

Observation periods + study period



Denominator population

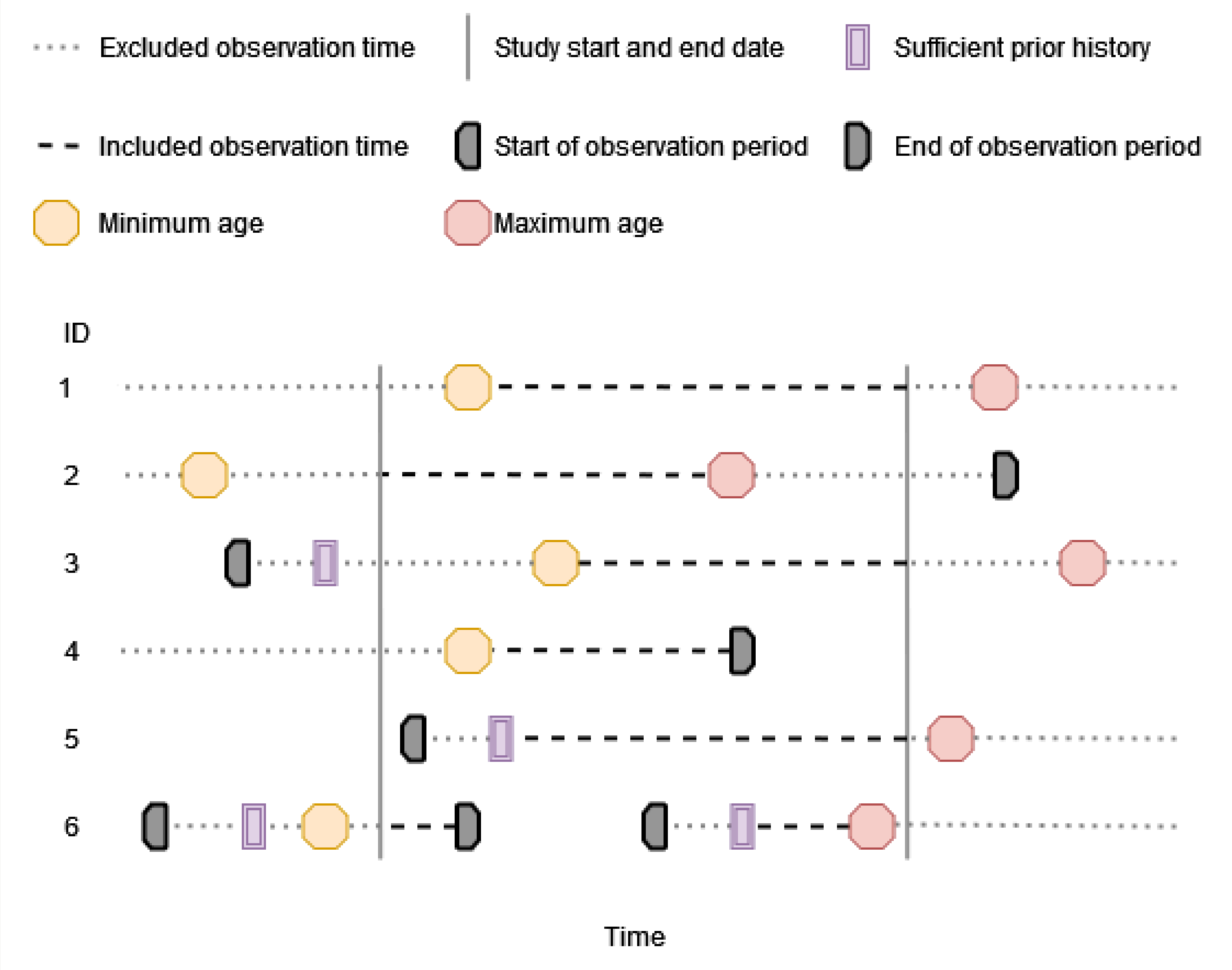
Observation periods + study period + prior history requirement



⬅ Back to Presentations

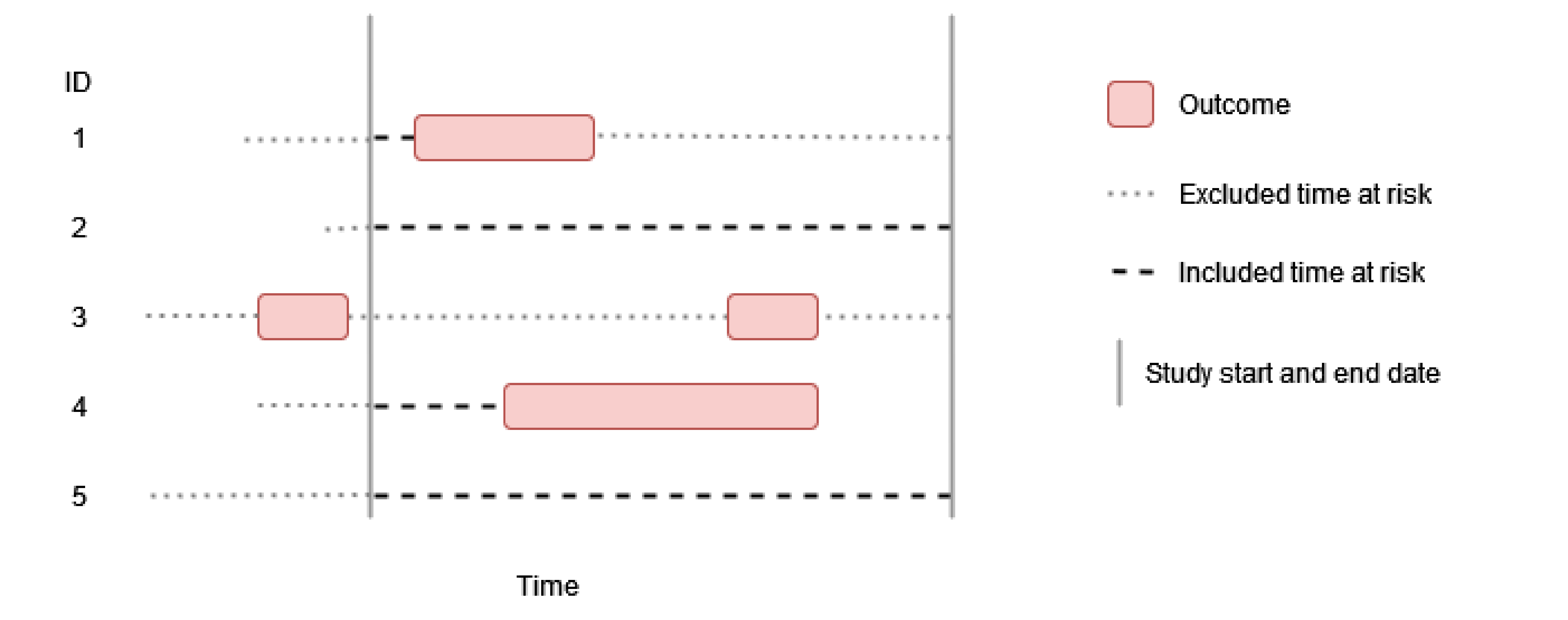
Denominator population

Observation periods + study period + prior history requirement + age (and sex) restriction



Incidence rates

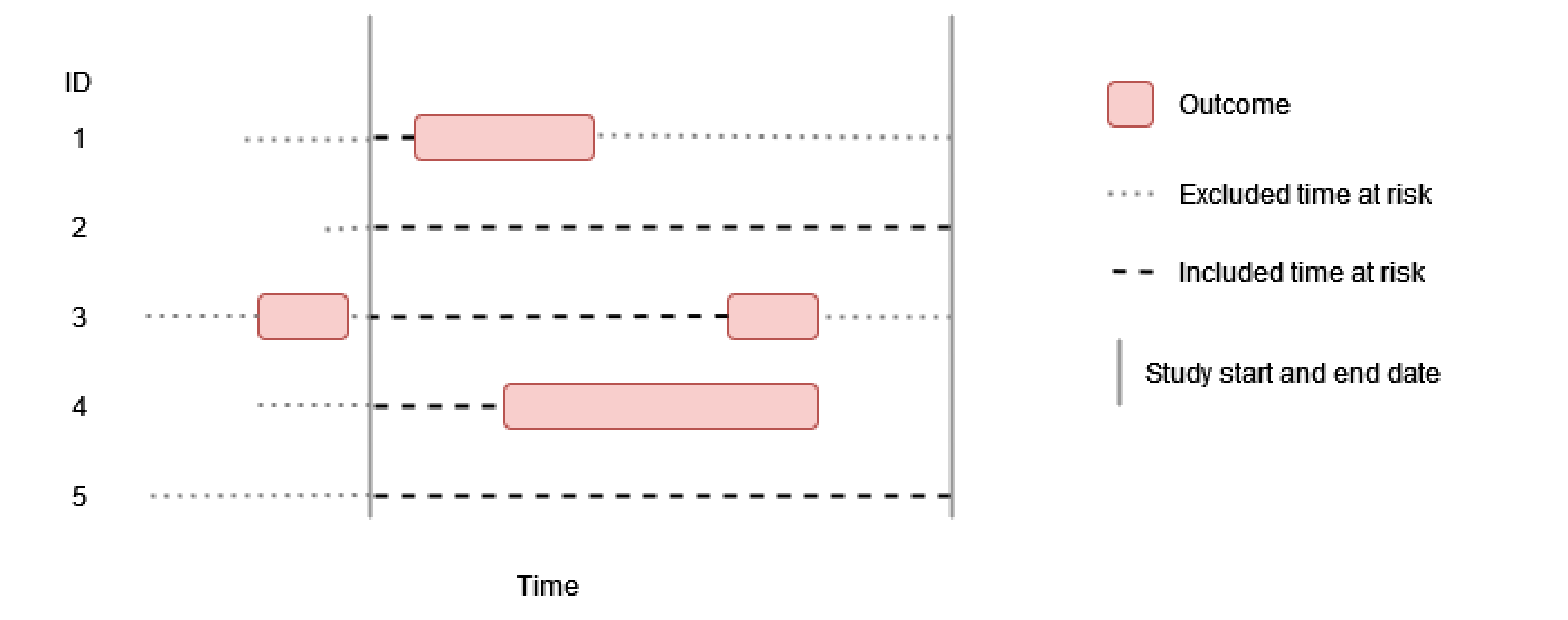
Washout all history, no repetitive events



⬅ Back to Presentations

Incidence rates

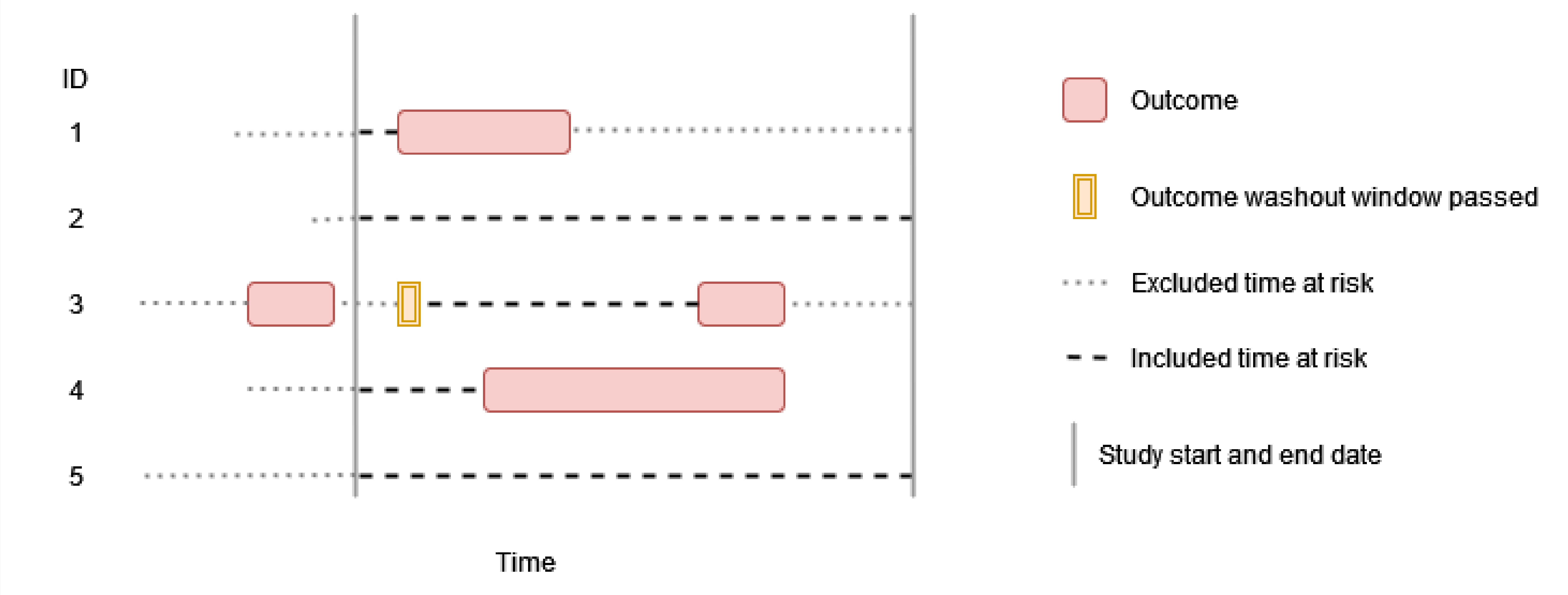
No washout, no repetitive events



⬅ Back to Presentations

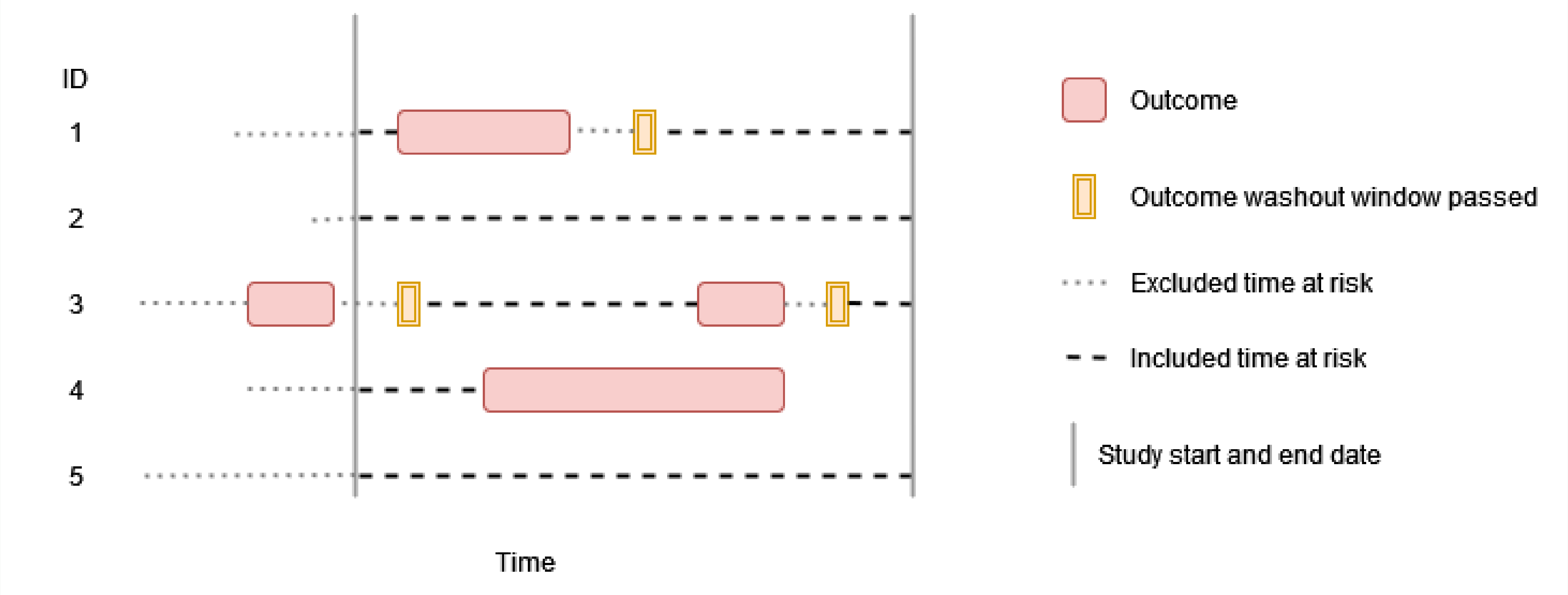
Incidence rates

Some washout, no repetitive events



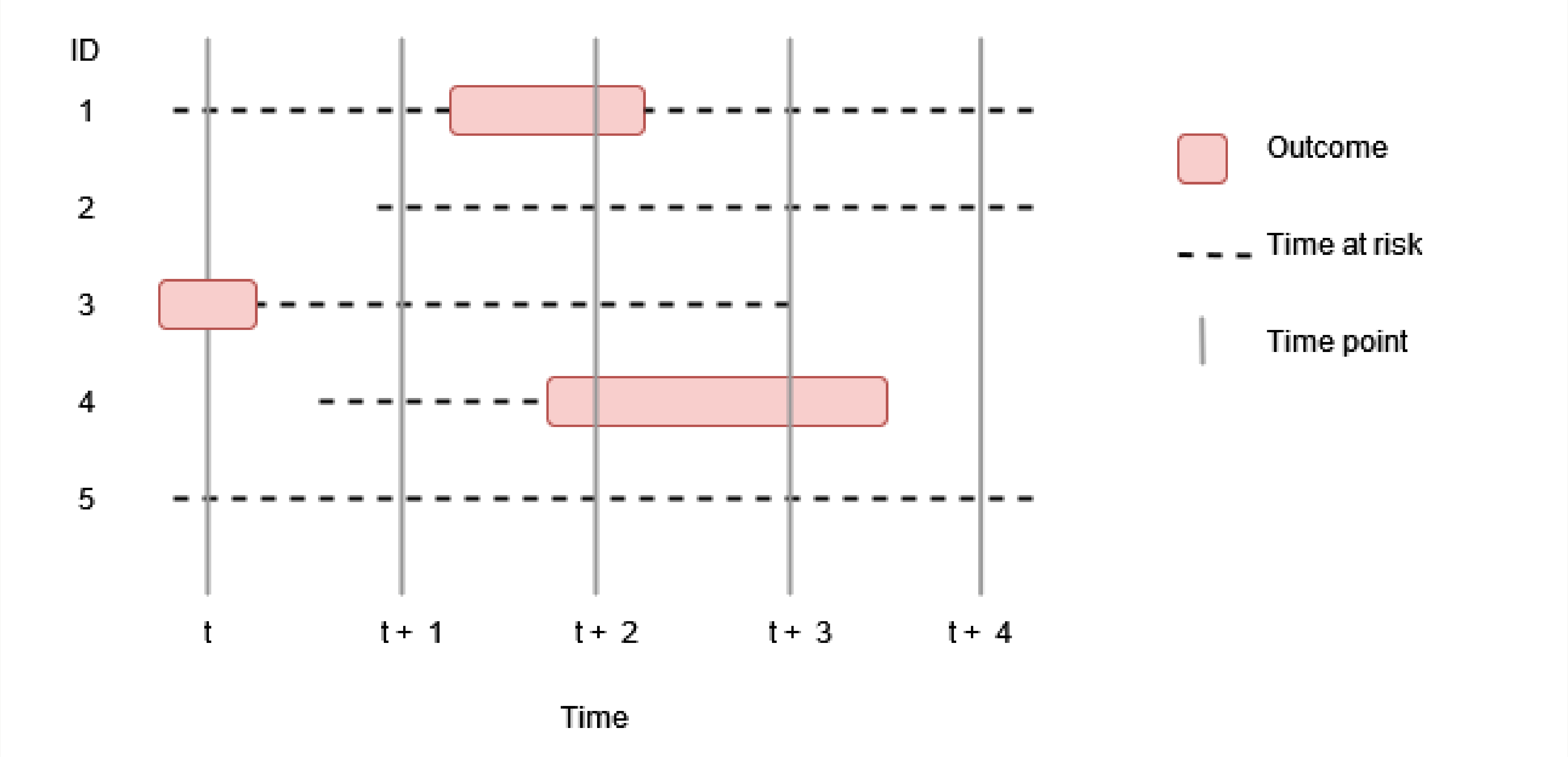
Incidence rates

Some washout, repetitive events



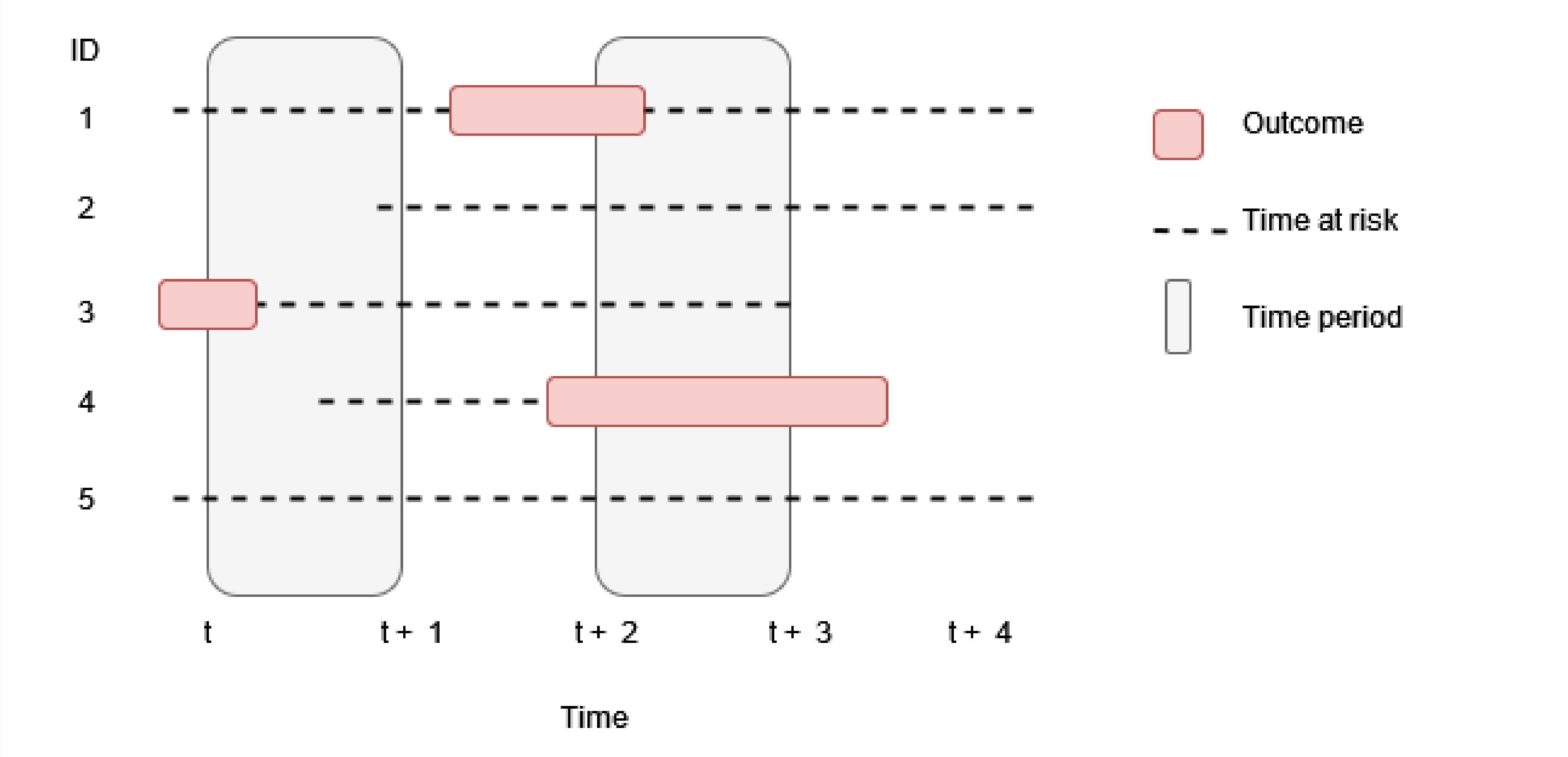
Prevalence

Point prevalence



Prevalence

Period prevalence



Oveview

- Concepts
- **Interface**
- More information

Required packages

```r
install.packages("IncidencePrevalence")
```

```r
library(CDMConnector)
library(IncidencePrevalence)
library(dplyr)
library(tidyr)
library(ggplot2)
library(gt)
```

generateDenominatorCohortSet()

```
cdm <- mockIncidencePrevalence(sampleSize = 50000)

cdm <- generateDenominatorCohortSet(cdm, name = "dpop")

cdm$dpop %>%
  glimpse()
```

```
Rows: ??
Columns: 4
$ cohort_definition_id <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,…
$ subject_id           <int> 1, 3, 7, 8, 11, 12, 14, 21, 22, 29, 32, 33, 36, 37, 41, 42, 43, 45, 46, 47, 49, 51, 52, 5…
$ cohort_start_date    <date> 1949-10-29, 1977-03-02, 1994-07-10, 1960-05-06, 1945-08-21, 1950-04-30, 1981-06-10, 1973…
$ cohort_end_date      <date> 1957-03-20, 1982-06-05, 1996-11-16, 1963-04-12, 1951-02-11, 1954-05-09, 1982-05-31, 1975…
```

generateDenominatorCohortSet()

```r
cdm <- generateDenominatorCohortSet(
  cdm = cdm, name = "dpop",
  cohortDateRange = as.Date(c("2008-01-01", "2012-01-01"))
)

cdm$dpop %>%
  glimpse()
```

```
Rows: ??
Columns: 4
$ cohort_definition_id <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,…
$ subject_id           <int> 33, 41, 52, 59, 69, 76, 79, 89, 107, 122, 126, 136, 141, 151, 156, 159, 164, 174, 175, 17…
$ cohort_start_date    <date> 2008-01-01, 2008-01-01, 2008-01-01, 2008-01-01, 2008-01-01, 2008-01-01, 2008-01-01, 2008…
$ cohort_end_date      <date> 2008-08-04, 2012-01-01, 2011-10-24, 2012-01-01, 2012-01-01, 2009-03-21, 2012-01-01, 2009…
```

generateDenominatorCohortSet()

```r
cohortCount(cdm$dpop)
```

```
# A tibble: 1 × 3
  cohort_definition_id number_records number_subjects
                 <int>          <int>           <int>
1                    1           3575            3575
```

```r
settings(cdm$dpop)
```

```
# A tibble: 1 × 11
  cohort_definition_id cohort_name    age_group sex   days_prior_observation start_date end_date   requirements_at_entry
                 <int> <chr>          <chr>     <chr>                  <dbl> <date>     <date>     <chr>
1                    1 denominator_c… 0 to 150  Both                       0 2008-01-01 2012-01-01 FALSE
# ℹ 3 more variables: target_cohort_definition_id <int>, target_cohort_name <chr>, time_at_risk <chr>
```

generateDenominatorCohortSet()

```
attrition(cdm$dpop)
```

```
# A tibble: 8 × 7
  cohort_definition_id number_records number_subjects reason_id reason                     excluded_records excluded_subjects
                 <int>          <int>           <int>     <int> <chr>                                 <int>             <int>
1                    1          50000           50000         1 Starting population                      NA                NA
2                    1          50000           50000         2 Missing year of birth                     0                 0
3                    1          50000           50000         3 Missing sex                               0                 0
4                    1          50000           50000         4 Cannot satisfy age c…                     0                 0
5                    1           3575            3575         5 No observation time …                 46425             46425
6                    1           3575            3575         6 Doesn't satisfy age …                     0                 0
7                    1           3575            3575         7 Prior history requir…                     0                 0
8                    1           3575            3575        10 No observation time …                     0                 0
```

generateDenominatorCohortSet()

```r
cdm <- generateDenominatorCohortSet(
  cdm = cdm, name = "dpop",
  cohortDateRange = as.Date(c("2008-01-01", "2012-01-01")),
  ageGroup = list(
    c(0, 49),
    c(50, 100)
  ),
  sex = c("Male", "Female")
)


cdm$dpop %>%
  glimpse()
```

```
Rows: ??
Columns: 4
$ cohort_definition_id <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,…
$ subject_id           <int> 59, 89, 136, 200, 208, 396, 501, 579, 581, 615, 642, 665, 740, 744, 773, 834, 910, 923, 1…
$ cohort_start_date    <date> 2008-01-01, 2008-01-01, 2008-01-01, 2008-01-01, 2008-01-01, 2009-08-30, 2008-01-01, 2008…
$ cohort_end_date      <date> 2012-01-01, 2009-08-06, 2010-02-18, 2010-12-23, 2012-01-01, 2012-01-01, 2008-12-05, 2011…
```

generateDenominatorCohortSet()

```r
cdm <- generateDenominatorCohortSet(
  cdm = cdm, name = "dpop",
  cohortDateRange = as.Date(c("2008-01-01", "2012-01-01")),
  ageGroup = list(
    c(0, 49),
    c(50, 100)
  ),
  sex = c("Male", "Female"),
  daysPriorObservation= c(0, 180)
)


cdm$dpop %>%
  glimpse()
```

```
Rows: ??
Columns: 4
$ cohort_definition_id <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,…
$ subject_id           <int> 59, 89, 136, 200, 208, 396, 501, 579, 581, 615, 642, 665, 740, 744, 773, 834, 910, 923, 1…
$ cohort_start_date    <date> 2008-01-01, 2008-01-01, 2008-01-01, 2008-01-01, 2008-01-01, 2009-08-30, 2008-01-01, 2008…
$ cohort_end_date      <date> 2012-01-01, 2009-08-06, 2010-02-18, 2010-12-23, 2012-01-01, 2012-01-01, 2008-12-05, 2011…
```

⬅ Back to Presentations

generateDenominatorCohortSet()

```r
settings(cdm$dpop)
```

```
# A tibble: 8 × 11
  cohort_definition_id cohort_name    age_group sex   days_prior_observation start_date end_date   requirements_at_entry
                 <int> <chr>          <chr>     <chr>                  <dbl> <date>     <date>     <chr>
1                    1 denominator_c… 0 to 49   Male                       0 2008-01-01 2012-01-01 FALSE
2                    2 denominator_c… 0 to 49   Male                     180 2008-01-01 2012-01-01 FALSE
3                    3 denominator_c… 0 to 49   Fema…                      0 2008-01-01 2012-01-01 FALSE
4                    4 denominator_c… 0 to 49   Fema…                    180 2008-01-01 2012-01-01 FALSE
5                    5 denominator_c… 50 to 100 Male                       0 2008-01-01 2012-01-01 FALSE
6                    6 denominator_c… 50 to 100 Male                     180 2008-01-01 2012-01-01 FALSE
7                    7 denominator_c… 50 to 100 Fema…                      0 2008-01-01 2012-01-01 FALSE
8                    8 denominator_c… 50 to 100 Fema…                    180 2008-01-01 2012-01-01 FALSE
# ℹ 3 more variables: target_cohort_definition_id <int>, target_cohort_name <chr>, time_at_risk <chr>
```

generateDenominatorCohortSet()

```r
cohortCount(cdm$dpop)
```

```
# A tibble: 8 × 3
  cohort_definition_id number_records number_subjects
                 <int>          <int>           <int>
1                    1            973             973
2                    2            959             959
3                    3            902             902
4                    4            890             890
5                    5            933             933
6                    6            925             925
7                    7            922             922
8                    8            900             900
```

generateDenominatorCohortSet()

```r
attrition(cdm$dpop) %>%
  filter(cohort_definition_id == 1)
```

```
# A tibble: 9 × 7
  cohort_definition_id number_records number_subjects reason_id reason                 excluded_records excluded_subjects
                 <int>          <int>           <int>     <int> <chr>                             <int>             <int>
1                    1          50000           50000         1 Starting population                  NA                NA
2                    1          50000           50000         2 Missing year of birth                 0                 0
3                    1          50000           50000         3 Missing sex                           0                 0
4                    1          50000           50000         4 Cannot satisfy age c…                 0                 0
5                    1           3575            3575         5 No observation time …             46425             46425
6                    1           3575            3575         6 Doesn't satisfy age …                 0                 0
7                    1           3575            3575         7 Prior history requir…                 0                 0
8                    1           1822            1822         8 Not Male                           1753              1753
9                    1            973             973        10 No observation time …               849               849
```

generateDenominatorCohortSet()

```r
# get some people who are in more than one cohort
ids <- cdm$dpop %>%
  group_by(subject_id) %>%
  tally() %>%
  collect() %>%
  arrange(desc(n)) %>%
  head(4) %>%
  pull("subject_id")
```

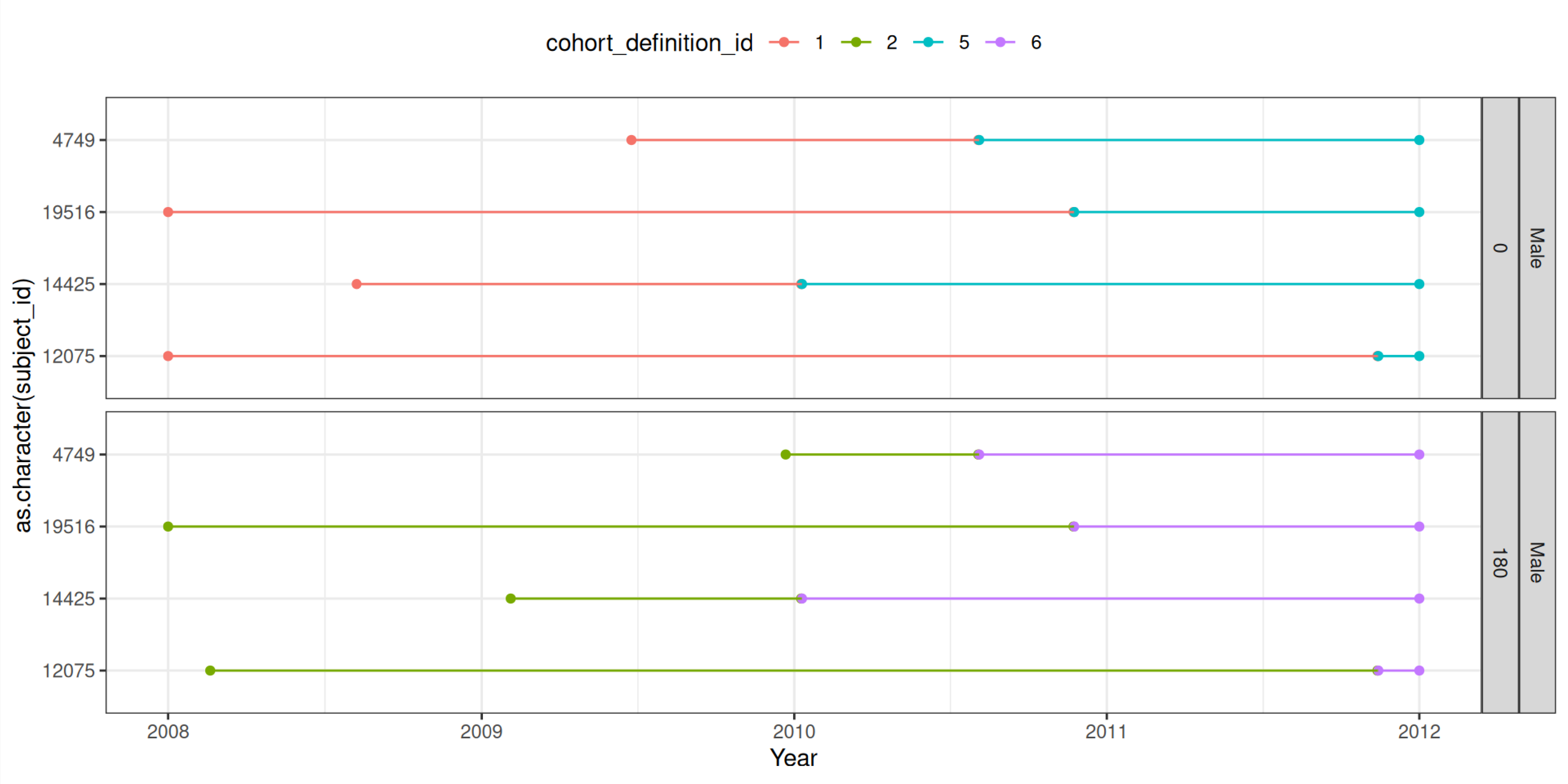
generateDenominatorCohortSet()

```
dpop <- cdm$dpop %>%
  collect() %>%
  left_join(settings(cdm$dpop),
            by = "cohort_definition_id") %>%
  mutate(cohort_definition_id=as.character(cohort_definition_id))

plot <- dpop %>%
  filter(subject_id %in% ids) %>%
  pivot_longer(cols = c(
    "cohort_start_date",
    "cohort_end_date"
  )) %>%
  ggplot(aes(x = as.character(subject_id), y = value, colour = cohort_definition_id, group = subject_id)) +
  facet_grid(sex + days_prior_observation ~ ., space = "free_y") +
  geom_point(position = position_dodge(width = 0.5)) +
  geom_line(position = position_dodge(width = 0.5)) +
  theme_bw() +
  theme(legend.position = "top") +
```

generateDenominatorCohortSet()

```
plot
```



Adding (time-invariant) variables for stratification

If later we want to estimate incidence or prevalence stratified for some time-invariant characteristic, we will need to add a variable to our denominator cohort table.

```
cdm$dpop <- cdm$dpop %>%
  mutate(group = if_else(as.numeric(subject_id)  < 20, "first", "second"))

cdm$dpop |>
  glimpse()
```

```
Rows: ??
Columns: 5
$ cohort_definition_id <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,…
$ subject_id           <int> 59, 89, 136, 200, 208, 396, 501, 579, 581, 615, 642, 665, 740, 744, 773, 834, 910, 923, 1…
$ cohort_start_date    <date> 2008-01-01, 2008-01-01, 2008-01-01, 2008-01-01, 2008-01-01, 2009-08-30, 2008-01-01, 2008…
$ cohort_end_date      <date> 2012-01-01, 2009-08-06, 2010-02-18, 2010-12-23, 2012-01-01, 2012-01-01, 2008-12-05, 2011…
$ group                <chr> "second", "second", "second", "second", "second", "second", "second", "second", "second",…
```

generateTargetDenominatorCohortSet()

When we want to stratify on a time-varying characteristic, we will do this by first creating a cohort for it. Once we have the cohort we will will use it when creating our denominator cohort.

⬅ Back to Presentations

generateTargetDenominatorCohortSet()

```r
personTable <- tibble(
  person_id = c("1", "2", "3", "4", "5"),
  gender_concept_id = c(rep("8507", 2), rep("8532", 3)),
  year_of_birth = 2000,
  month_of_birth = 06,
  day_of_birth = 01
)
observationPeriodTable <- tibble(
  observation_period_id = "1",
  person_id = c("1", "2", "3", "4", "5"),
  observation_period_start_date = c(
    as.Date("2010-12-19"),
    as.Date("2005-04-01"),
    as.Date("2009-04-10"),
    as.Date("2010-08-20"),
    as.Date("2010-01-01")
  ),
  observation_period_end_date = c(
```

⬅ Back to Presentations

generateTargetDenominatorCohortSet()

```
cdm <- generateTargetDenominatorCohortSet(
  cdm = cdm,
  name = "denominator_acute_asthma",
  targetCohortTable = "target"
)

cdm$denominator_acute_asthma |>
  dplyr::glimpse()
```

```
Rows: ??
Columns: 4
$ cohort_definition_id <int> 1, 1, 1, 1, 1
$ subject_id           <chr> "2", "3", "5", "3", "5"
$ cohort_start_date    <date> 2005-08-20, 2015-06-01, 2010-06-01, 2012-01-01, 2014-10-01
$ cohort_end_date      <date> 2005-09-20, 2015-12-31, 2010-06-01, 2013-01-01, 2015-04-01
```

generateTargetDenominatorCohortSet()

We can add demographic requirements like before. But it is important to note that these are applied at the cohort start date of the target cohort.

```
cdm <- generateTargetDenominatorCohortSet(
  cdm = cdm,
  name = "denominator_acute_asthma_2",
  ageGroup = list(c(11, 15)),
  sex = "Female",
  daysPriorObservation = 0,
  targetCohortTable = "target"
)
cdm$denominator_acute_asthma_2 |>
  dplyr::glimpse()
```

```
Rows: ??
Columns: 4
$ cohort_definition_id <int> 1, 1, 1
$ subject_id           <chr> "3", "3", "5"
$ cohort_start_date    <date> 2015-06-01, 2012-01-01, 2014-10-01
$ cohort_end_date      <date> 2015-12-31, 2013-01-01, 2015-04-01
```

estimateIncidence()

```r
cdm <- mockIncidencePrevalence(
  sampleSize = 50000,
  outPre = 0.5
)

cdm <- generateDenominatorCohortSet(
  cdm = cdm, name = "denominator",
  cohortDateRange = as.Date(c("2008-01-01", "2012-01-01")),
  ageGroup = list(
    c(0, 30),
    c(31, 50),
    c(51, 70),
    c(71, 100)
  )
)
inc <- estimateIncidence(
  cdm = cdm,
  denominatorTable = "denominator"
```

estimateIncidence()

```r
inc %>%
  glimpse()
```

```
Rows: 288
Columns: 13
$ result_id        <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, …
$ cdm_name         <chr> "mock", "mock", "mock", "mock", "mock", "mock", "mock", "mock", "mock", "mock", "mock", "mock…
$ group_name       <chr> "denominator_cohort_name &&& outcome_cohort_name", "denominator_cohort_name &&& outcome_cohor…
$ group_level      <chr> "denominator_cohort_1 &&& cohort_1", "denominator_cohort_1 &&& cohort_1", "denominator_cohort…
$ strata_name      <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over…
$ strata_level     <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over…
$ variable_name    <chr> "Denominator", "Outcome", "Denominator", "Denominator", "Outcome", "Outcome", "Outcome", "Den…
$ variable_level   <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N…
$ estimate_name    <chr> "denominator_count", "outcome_count", "person_days", "person_years", "incidence_100000_pys", …
$ estimate_type    <chr> "integer", "integer", "numeric", "numeric", "numeric", "numeric", "numeric", "integer", "inte…
$ estimate_value   <chr> "662", "63", "189075", "517.659", "12170.174", "9351.896", "15570.939", "624", "64", "180488"…
$ additional_name  <chr> "incidence_start_date &&& incidence_end_date &&& analysis_interval", "incidence_start_date &&…
```

estimateIncidence()

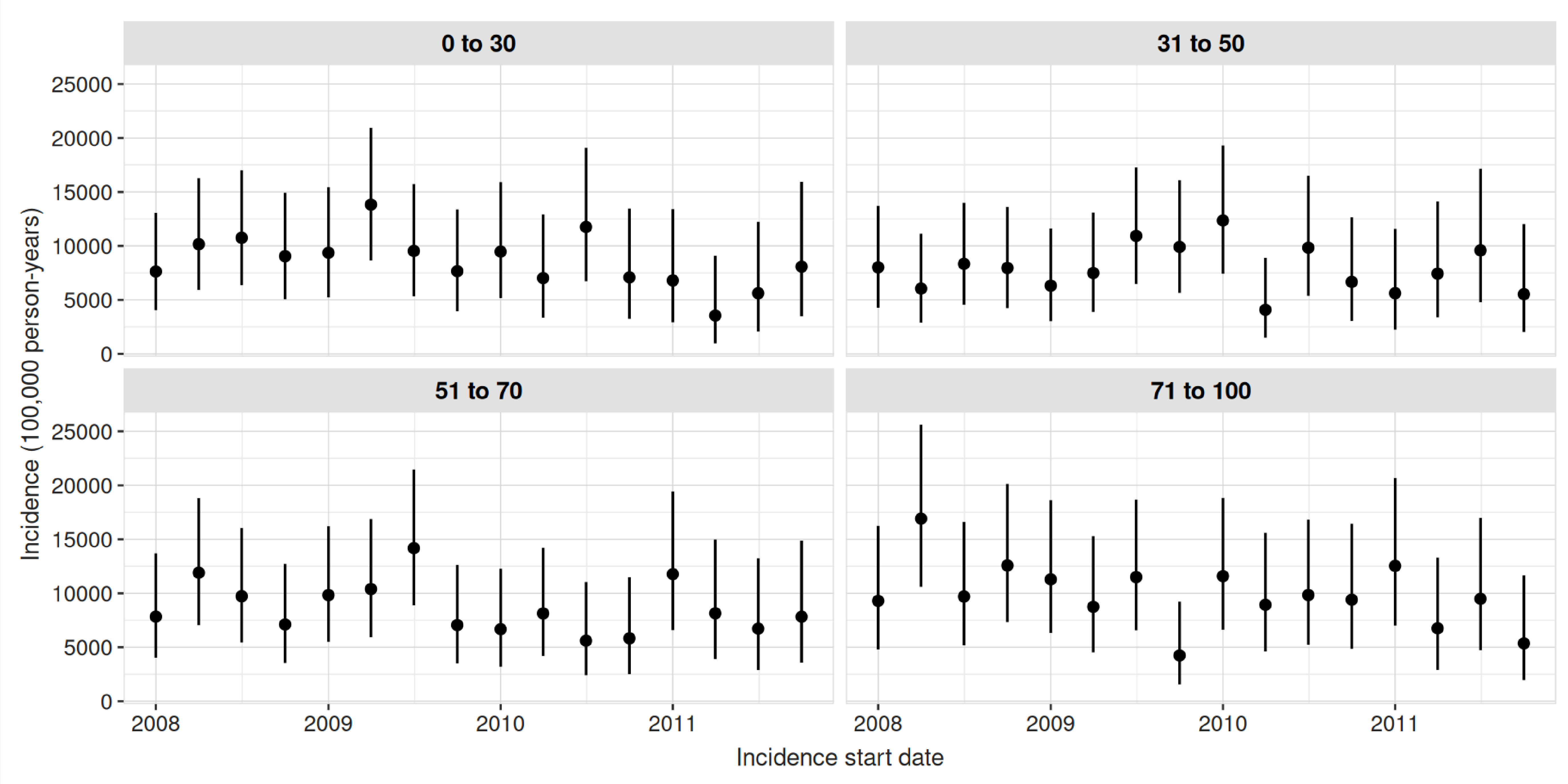
```
inc <- estimateIncidence(
  cdm = cdm,
  denominatorTable = "denominator",
  outcomeTable = "outcome",
  interval = "quarters",
  outcomeWashout = 365,
  repeatedEvents = TRUE
)
inc %>%
  glimpse()
```

```
Rows: 624
Columns: 13
$ result_id        <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, …
$ cdm_name         <chr> "mock", "mock", "mock", "mock", "mock", "mock", "mock", "mock", "mock", "mock", "mock", "mock…
$ group_name       <chr> "denominator_cohort_name &&& outcome_cohort_name", "denominator_cohort_name &&& outcome_cohor…
$ group_level      <chr> "denominator_cohort_1 &&& cohort_1", "denominator_cohort_1 &&& cohort_1", "denominator_cohort…
$ strata_name      <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over…
$ strata_level     <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over…
$ variable_name    <chr> "Denominator", "Outcome", "Denominator", "Denominator", "Outcome", "Outcome", "Outcome", "Den…
$ variable_level   <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N…
$ estimate_name    <chr> "denominator_count", "outcome_count", "person_days", "person_years", "incidence_100000_pys", …
$ estimate_type    <chr> "integer", "integer", "numeric", "numeric", "numeric", "numeric", "numeric", "integer", "inte…
$ estimate_value   <chr> "732", "13", "62234", "170.387", "7629.69", "4062.489", "13047.002", "718", "17", "61093", "1…
$ additional_name  <chr> "incidence_start_date &&& incidence_end_date &&& analysis_interval", "incidence_start_date &&…
```

⬅ Back to Presentations

plotIncidence()

```r
plotIncidence(inc,
              facet = "denominator_age_group")
```



⬅ Back to Presentations

estimateIncidence()

```
tableIncidence(inc)
```

Incidence start date	Incidence end date	Denominator age group	Denominator sex	Estimate name			
				Denominator (N)	Person-years	Outcome (N)	Incidence 100,000 person-years [95% CI]
mock; cohort_1							
2008-01-01	2008-03-31	0 to 30	Both	732	170.39	13	7,629.69 (4,062.49 - 13,047.00)
2008-04-01	2008-06-30	0 to 30	Both	718	167.26	17	10,163.57 (5,920.66 - 16,272.87)
2008-07-01	2008-09-30	0 to 30	Both	708	167.39	18	10,753.52 (6,373.22 - 16,995.20)
2008-10-01	2008-12-31	0 to 30	Both	715	165.77	15	9,048.63 (5,064.45 - 14,924.33)
2009-01-01	2009-03-31	0 to 30	Both	695	160.15	15	9,366.51 (5,242.36 - 15,448.64)
2009-04-01	2009-06-30	0 to 30	Both	695	159.13	22	13,825.43 (8,664.33 - 20,931.88)
2009-07-01	2009-09-30	0 to 30	Both	670	157.35	15	9,533.07 (5,335.59 - 15,723.35)
2009-10-01	2009-12-31	0 to 30	Both	656	156.67	12	7,659.22 (3,957.63 - 13,379.11)
2010-01-01	2010-03-31	0 to 30	Both	626	147.78	14	9,473.28 (5,179.13 - 15,894.56)
2010-04-01	2010-06-30	0 to 30	Both	600	142.42	10	7,021.34 (3,367.00 - 12,912.49)
2010-07-01	2010-09-30	0 to 30	Both	570	136.14	16	11,752.61 (6,717.63 - 19,085.50)
2010-10-01	2010-12-31	0 to 30	Both	530	126.99	9	7,087.17 (3,240.71 - 13,453.66)

Incidence start date	Incidence end date	Denominator age group	Denominator sex	Estimate name			
				Denominator (N)	Person-years	Outcome (N)	Incidence 100,000 person-years [95% CI]
2011-01-01	2011-03-31	0 to 30	Both	497	117.62	8	6,801.33 (2,936.33 - 13,401.34)
2011-04-01	2011-06-30	0 to 30	Both	470	112.61	4	3,551.99 (967.80 - 9,094.50)
2011-07-01	2011-09-30	0 to 30	Both	447	106.78	6	5,619.08 (2,062.10 - 12,230.38)
2011-10-01	2011-12-31	0 to 30	Both	418	99.00	8	8,080.97 (3,488.79 - 15,922.74)
2008-01-01	2008-03-31	31 to 50	Both	694	162.12	13	8,018.95 (4,269.75 - 13,712.65)
2008-04-01	2008-06-30	31 to 50	Both	702	165.28	10	6,050.19 (2,901.30 - 11,126.52)
2008-07-01	2008-09-30	31 to 50	Both	712	167.91	14	8,337.70 (4,558.30 - 13,989.25)
2008-10-01	2008-12-31	31 to 50	Both	695	163.34	13	7,959.10 (4,237.89 - 13,610.31)
2009-01-01	2009-03-31	31 to 50	Both	682	158.32	10	6,316.24 (3,028.88 - 11,615.79)
2009-04-01	2009-06-30	31 to 50	Both	689	160.18	12	7,491.67 (3,871.05 - 13,086.43)
2009-07-01	2009-09-30	31 to 50	Both	699	164.70	18	10,929.09 (6,477.27 - 17,272.68)
2009-10-01	2009-12-31	31 to 50	Both	697	161.48	16	9,908.29 (5,663.44 - 16,090.44)
2010-01-01	2010-03-31	31 to 50	Both	655	153.79	19	12,354.27 (7,438.08 - 19,292.72)
2010-04-01	2010-06-30	31 to 50	Both	621	146.93	6	4,083.58 (1,498.60 - 8,888.23)
2010-07-01	2010-09-30	31 to 50	Both	589	142.36	14	9,834.02 (5,376.35 - 16,499.81)
2010-10-01	2010-12-31	31 to 50	Both	568	135.00	9	6,666.76 (3,048.47 - 12,655.60)

[⬅ Back to Presentations]

Incidence start date	Incidence end date	Denominator age group	Denominator sex	Estimate name			
				Denominator (N)	Person-years	Outcome (N)	Incidence 100,000 person-years [95% CI]
2011-01-01	2011-03-31	31 to 50	Both	535	124.54	7	5,620.46 (2,259.72 - 11,580.29)
2011-04-01	2011-06-30	31 to 50	Both	504	121.08	9	7,432.86 (3,398.78 - 14,109.88)
2011-07-01	2011-09-30	31 to 50	Both	486	114.72	11	9,588.90 (4,786.74 - 17,157.19)
2011-10-01	2011-12-31	31 to 50	Both	455	108.55	6	5,527.51 (2,028.50 - 12,031.06)
2008-01-01	2008-03-31	51 to 70	Both	662	153.18	12	7,834.18 (4,048.03 - 13,684.73)
2008-04-01	2008-06-30	51 to 70	Both	655	151.24	18	11,901.69 (7,053.70 - 18,809.80)
2008-07-01	2008-09-30	51 to 70	Both	656	154.21	15	9,727.06 (5,444.16 - 16,043.30)
2008-10-01	2008-12-31	51 to 70	Both	646	154.67	11	7,112.05 (3,550.31 - 12,725.43)
2009-01-01	2009-03-31	51 to 70	Both	658	152.59	15	9,830.52 (5,502.07 - 16,213.95)
2009-04-01	2009-06-30	51 to 70	Both	666	153.96	16	10,392.44 (5,940.18 - 16,876.68)
2009-07-01	2009-09-30	51 to 70	Both	667	155.18	22	14,176.99 (8,884.65 - 21,464.14)
2009-10-01	2009-12-31	51 to 70	Both	660	155.92	11	7,054.76 (3,521.71 - 12,622.92)
2010-01-01	2010-03-31	51 to 70	Both	633	149.84	10	6,673.92 (3,200.40 - 12,273.58)
2010-04-01	2010-06-30	51 to 70	Both	619	147.54	12	8,133.39 (4,202.64 - 14,207.39)
2010-07-01	2010-09-30	51 to 70	Both	598	142.71	8	5,605.77 (2,420.18 - 11,045.61)
2010-10-01	2010-12-31	51 to 70	Both	574	137.31	8	5,826.27 (2,515.37 - 11,480.08)

⬅ Back to Presentations

Incidence start date	Incidence end date	Denominator age group	Denominator sex	Estimate name			
				Denominator (N)	Person-years	Outcome (N)	Incidence 100,000 person-years [95% CI]
2011-01-01	2011-03-31	51 to 70	Both	547	127.41	15	11,773.39 (6,589.48 - 19,418.41)
2011-04-01	2011-06-30	51 to 70	Both	525	122.76	10	8,145.98 (3,906.31 - 14,980.74)
2011-07-01	2011-09-30	51 to 70	Both	498	119.00	8	6,722.97 (2,902.50 - 13,246.93)
2011-10-01	2011-12-31	51 to 70	Both	482	114.96	9	7,829.01 (3,579.92 - 14,861.91)
2008-01-01	2008-03-31	71 to 100	Both	553	129.06	12	9,298.00 (4,804.41 - 16,241.74)
2008-04-01	2008-06-30	71 to 100	Both	568	130.07	22	16,913.58 (10,599.65 - 25,607.36)
2008-07-01	2008-09-30	71 to 100	Both	565	133.99	13	9,702.00 (5,165.91 - 16,590.71)
2008-10-01	2008-12-31	71 to 100	Both	572	135.16	17	12,577.22 (7,326.69 - 20,137.35)
2009-01-01	2009-03-31	71 to 100	Both	581	132.86	15	11,289.91 (6,318.88 - 18,620.99)
2009-04-01	2009-06-30	71 to 100	Both	585	137.14	12	8,750.25 (4,521.38 - 15,284.92)
2009-07-01	2009-09-30	71 to 100	Both	586	139.15	16	11,498.71 (6,572.51 - 18,673.19)
2009-10-01	2009-12-31	71 to 100	Both	610	141.44	6	4,242.23 (1,556.82 - 9,233.55)
2010-01-01	2010-03-31	71 to 100	Both	586	138.06	16	11,589.25 (6,624.26 - 18,820.21)
2010-04-01	2010-06-30	71 to 100	Both	563	134.34	12	8,932.29 (4,615.45 - 15,602.92)
2010-07-01	2010-09-30	71 to 100	Both	553	132.16	13	9,836.64 (5,237.59 - 16,820.95)
2010-10-01	2010-12-31	71 to 100	Both	522	127.57	12	9,406.38 (4,860.41 - 16,431.05)

⬅ Back to Presentations

estimatePointPrevalence() and estimatePeriodPrevalence()

```r
cdm <- mockIncidencePrevalence(
  sampleSize = 50000,
  outPre = 0.5
)

cdm <- generateDenominatorCohortSet(
  cdm = cdm, name = "denominator",
  cohortDateRange = as.Date(c("2008-01-01", "2012-01-01")),
  ageGroup = list(
    c(0, 30),
    c(31, 50),
    c(51, 70),
    c(71, 100)
  )
)
prev <- estimatePointPrevalence(
  cdm = cdm,
  denominatorTable = "denominator"
```

estimatePointPrevalence() and estimatePeriodPrevalence()

```r
prev %>%
  glimpse()
```

```
Rows: 276
Columns: 13
$ result_id         <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, …
$ cdm_name          <chr> "mock", "mock", "mock", "mock", "mock", "mock", "mock", "mock", "mock", "mock", "mock", "mock…
$ group_name        <chr> "denominator_cohort_name &&& outcome_cohort_name", "denominator_cohort_name &&& outcome_cohor…
$ group_level       <chr> "denominator_cohort_1 &&& cohort_1", "denominator_cohort_1 &&& cohort_1", "denominator_cohort…
$ strata_name       <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over…
$ strata_level      <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over…
$ variable_name     <chr> "Denominator", "Outcome", "Outcome", "Outcome", "Outcome", "Denominator", "Outcome", "Outcome…
$ variable_level    <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N…
$ estimate_name     <chr> "denominator_count", "outcome_count", "prevalence", "prevalence_95CI_lower", "prevalence_95CI…
$ estimate_type     <chr> "integer", "integer", "numeric", "numeric", "numeric", "integer", "integer", "numeric", "nume…
$ estimate_value    <chr> "755", "1", "0.00132", "0.00023", "0.00746", "708", "1", "0.00141", "0.00025", "0.00796", "67…
$ additional_name   <chr> "prevalence_start_date &&& prevalence_end_date &&& analysis_interval", "prevalence_start_date…
```

⬅ Back to Presentations

estimatePointPrevalence() and estimatePeriodPrevalence()

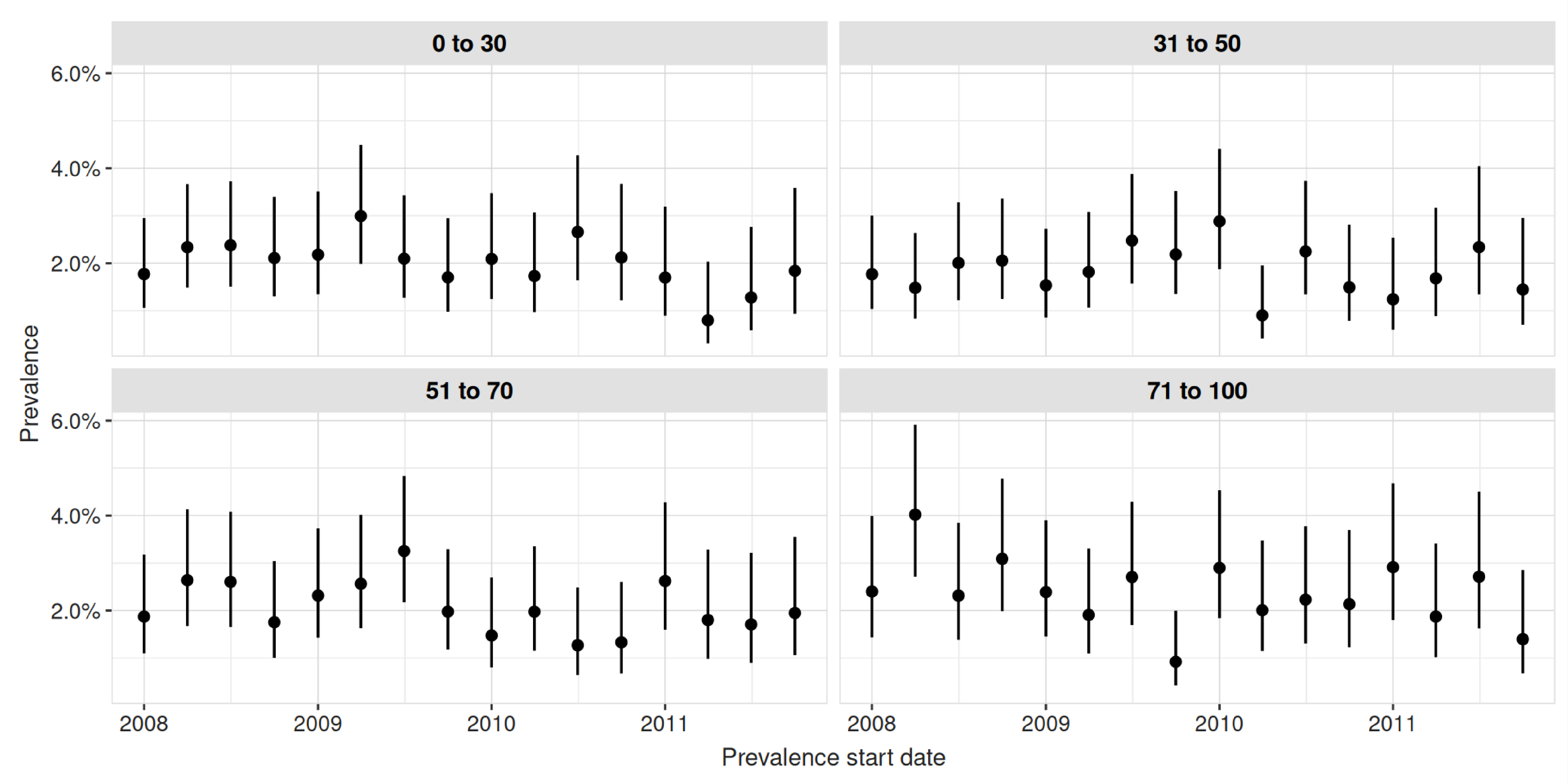
```r
prev <- estimatePeriodPrevalence(
  cdm = cdm,
  denominatorTable = "denominator",
  outcomeTable = "outcome",
  interval = "quarters"
)

prev %>%
  glimpse()
```

```
Rows: 496
Columns: 13
$ result_id        <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, …
$ cdm_name         <chr> "mock", "mock", "mock", "mock", "mock", "mock", "mock", "mock", "mock", "mock", "mock", "mock…
$ group_name       <chr> "denominator_cohort_name &&& outcome_cohort_name", "denominator_cohort_name &&& outcome_cohor…
$ group_level      <chr> "denominator_cohort_1 &&& cohort_1", "denominator_cohort_1 &&& cohort_1", "denominator_cohort…
$ strata_name      <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over…
$ strata_level     <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over…
$ variable_name    <chr> "Denominator", "Outcome", "Outcome", "Outcome", "Outcome", "Denominator", "Outcome", "Outcome…
$ variable_level   <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N…
$ estimate_name    <chr> "denominator_count", "outcome_count", "prevalence", "prevalence_95CI_lower", "prevalence_95CI…
$ estimate_type    <chr> "integer", "integer", "numeric", "numeric", "numeric", "integer", "integer", "numeric", "nume…
$ estimate_value   <chr> "790", "14", "0.01772", "0.01059", "0.02953", "770", "18", "0.02338", "0.01484", "0.03665", "…
$ additional_name  <chr> "prevalence_start_date &&& prevalence_end_date &&& analysis_interval", "prevalence_start_date…
```

plotPrevalence()

```r
plotPrevalence(prev,
               facet = "denominator_age_group")
```



plotPrevalence()

```
tablePrevalence(prev)
```

Prevalence start date	Prevalence end date	Denominator age group	Denominator sex	Estimate name		
				Denominator (N)	Outcome (N)	Prevalence [95% CI]
mock; cohort_1						
2008-01-01	2008-03-31	0 to 30	Both	790	14	0.02 (0.01 - 0.03)
2008-04-01	2008-06-30	0 to 30	Both	770	18	0.02 (0.01 - 0.04)
2008-07-01	2008-09-30	0 to 30	Both	757	18	0.02 (0.02 - 0.04)
2008-10-01	2008-12-31	0 to 30	Both	759	16	0.02 (0.01 - 0.03)
2009-01-01	2009-03-31	0 to 30	Both	734	16	0.02 (0.01 - 0.04)
2009-04-01	2009-06-30	0 to 30	Both	735	22	0.03 (0.02 - 0.04)
2009-07-01	2009-09-30	0 to 30	Both	716	15	0.02 (0.01 - 0.03)
2009-10-01	2009-12-31	0 to 30	Both	705	12	0.02 (0.01 - 0.03)
2010-01-01	2010-03-31	0 to 30	Both	670	14	0.02 (0.01 - 0.03)
2010-04-01	2010-06-30	0 to 30	Both	636	11	0.02 (0.01 - 0.03)
2010-07-01	2010-09-30	0 to 30	Both	602	16	0.03 (0.02 - 0.04)
2010-10-01	2010-12-31	0 to 30	Both	566	12	0.02 (0.01 - 0.04)
2011-01-01	2011-03-31	0 to 30	Both	530	9	0.02 (0.01 - 0.03)
2011-04-01	2011-06-30	0 to 30	Both	501	4	0.01 (0.00 - 0.02)
2011-07-01	2011-09-30	0 to 30	Both	469	6	0.01 (0.01 - 0.03)
2011-10-01	2011-12-31	0 to 30	Both	435	8	0.02 (0.01 - 0.04)
2008-01-01	2008-03-31	31 to 50	Both	735	13	0.02 (0.01 - 0.03)
2008-04-01	2008-06-30	31 to 50	Both	742	11	0.01 (0.01 - 0.03)
2008-07-01	2008-09-30	31 to 50	Both	748	15	0.02 (0.01 - 0.03)

⬅ Back to Presentations

Prevalence start date	Prevalence end date	Denominator age group	Denominator sex	Estimate name		
				Denominator (N)	Outcome (N)	Prevalence [95% CI]
2008-10-01	2008-12-31	31 to 50	Both	730	15	0.02 (0.01 - 0.03)
2009-01-01	2009-03-31	31 to 50	Both	717	11	0.02 (0.01 - 0.03)
2009-04-01	2009-06-30	31 to 50	Both	716	13	0.02 (0.01 - 0.03)
2009-07-01	2009-09-30	31 to 50	Both	727	18	0.02 (0.02 - 0.04)
2009-10-01	2009-12-31	31 to 50	Both	732	16	0.02 (0.01 - 0.04)
2010-01-01	2010-03-31	31 to 50	Both	694	20	0.03 (0.02 - 0.04)
2010-04-01	2010-06-30	31 to 50	Both	665	6	0.01 (0.00 - 0.02)
2010-07-01	2010-09-30	31 to 50	Both	623	14	0.02 (0.01 - 0.04)
2010-10-01	2010-12-31	31 to 50	Both	603	9	0.01 (0.01 - 0.03)
2011-01-01	2011-03-31	31 to 50	Both	564	7	0.01 (0.01 - 0.03)
2011-04-01	2011-06-30	31 to 50	Both	535	9	0.02 (0.01 - 0.03)
2011-07-01	2011-09-30	31 to 50	Both	513	12	0.02 (0.01 - 0.04)
2011-10-01	2011-12-31	31 to 50	Both	484	7	0.01 (0.01 - 0.03)
2008-01-01	2008-03-31	51 to 70	Both	694	13	0.02 (0.01 - 0.03)
2008-04-01	2008-06-30	51 to 70	Both	682	18	0.03 (0.02 - 0.04)
2008-07-01	2008-09-30	51 to 70	Both	691	18	0.03 (0.02 - 0.04)
2008-10-01	2008-12-31	51 to 70	Both	684	12	0.02 (0.01 - 0.03)
2009-01-01	2009-03-31	51 to 70	Both	691	16	0.02 (0.01 - 0.04)
2009-04-01	2009-06-30	51 to 70	Both	702	18	0.03 (0.02 - 0.04)
2009-07-01	2009-09-30	51 to 70	Both	707	23	0.03 (0.02 - 0.05)
2009-10-01	2009-12-31	51 to 70	Both	708	14	0.02 (0.01 - 0.03)
2010-01-01	2010-03-31	51 to 70	Both	678	10	0.01 (0.01 - 0.03)
2010-04-01	2010-06-30	51 to 70	Both	658	13	0.02 (0.01 - 0.03)
2010-07-01	2010-09-30	51 to 70	Both	630	8	0.01 (0.01 - 0.02)
2010-10-01	2010-12-31	51 to 70	Both	601	8	0.01 (0.01 - 0.03)

Prevalence start date	Prevalence end date	Denominator age group	Denominator sex	Estimate name		
				Denominator (N)	Outcome (N)	Prevalence [95% CI]
2011-01-01	2011-03-31	51 to 70	Both	572	15	0.03 (0.02 - 0.04)
2011-04-01	2011-06-30	51 to 70	Both	555	10	0.02 (0.01 - 0.03)
2011-07-01	2011-09-30	51 to 70	Both	527	9	0.02 (0.01 - 0.03)
2011-10-01	2011-12-31	51 to 70	Both	513	10	0.02 (0.01 - 0.04)
2008-01-01	2008-03-31	71 to 100	Both	583	14	0.02 (0.01 - 0.04)
2008-04-01	2008-06-30	71 to 100	Both	597	24	0.04 (0.03 - 0.06)
2008-07-01	2008-09-30	71 to 100	Both	605	14	0.02 (0.01 - 0.04)
2008-10-01	2008-12-31	71 to 100	Both	615	19	0.03 (0.02 - 0.05)
2009-01-01	2009-03-31	71 to 100	Both	628	15	0.02 (0.01 - 0.04)
2009-04-01	2009-06-30	71 to 100	Both	629	12	0.02 (0.01 - 0.03)
2009-07-01	2009-09-30	71 to 100	Both	628	17	0.03 (0.02 - 0.04)
2009-10-01	2009-12-31	71 to 100	Both	651	6	0.01 (0.00 - 0.02)
2010-01-01	2010-03-31	71 to 100	Both	621	18	0.03 (0.02 - 0.05)
2010-04-01	2010-06-30	71 to 100	Both	598	12	0.02 (0.01 - 0.03)
2010-07-01	2010-09-30	71 to 100	Both	583	13	0.02 (0.01 - 0.04)
2010-10-01	2010-12-31	71 to 100	Both	562	12	0.02 (0.01 - 0.04)
2011-01-01	2011-03-31	71 to 100	Both	549	16	0.03 (0.02 - 0.05)
2011-04-01	2011-06-30	71 to 100	Both	534	10	0.02 (0.01 - 0.03)
2011-07-01	2011-09-30	71 to 100	Both	516	14	0.03 (0.02 - 0.05)
2011-10-01	2011-12-31	71 to 100	Both	501	7	0.01 (0.01 - 0.03)

Oveview

- Concepts
- Interface
- **More information**

⬅ Back to Presentations

Package paper

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ORIGINAL ARTICLE

IncidencePrevalence: An R package to calculate population-level incidence rates and prevalence using the OMOP common data model

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IncidencePrevalence

👉 **Packages website**
👉 **CRAN link**
👉 **Manual**

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